MILITARY SPECIFICATION SHEET

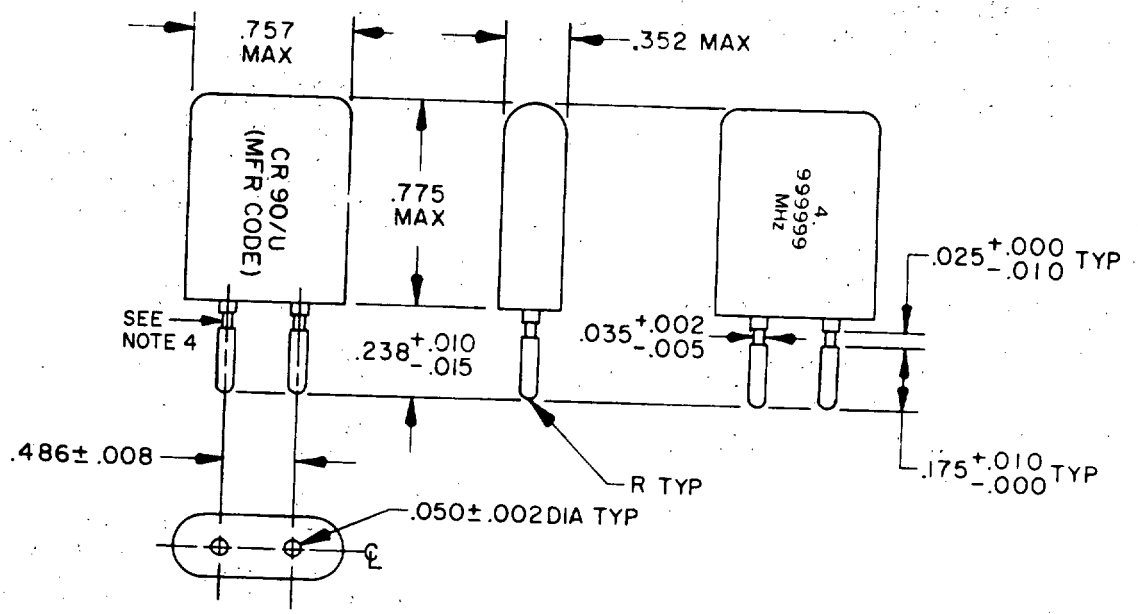
CRYSTAL UNIT, QUARTZ, CR-90/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Ⓒ **INACTIVE FOR NEW DESIGN**

Pertinent characteristics: 3 to 5 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type designation, frequency, and code designation characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. The pin undercut may be omitted.

FIGURE 1. Crtstal unit - CR-90/U.

Ⓒ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-27/U (MIL-H-10056/14)
Dimensions, marking, and configuration	See figure 1
Frequency range	3 to 5 MHz, inclusive
Frequency tolerance	±4 ppm at 30°C ± 0.5°C
Equivalent resistance	3 MHz 15 ohms, maximum 5 MHz 9 ohms, maximum
Antiresonance load capacitance	30 ±0.5 pF
Mode of oscillation	Fundamental
Operating temperature range	-55°C to +90°C, inclusive
Test set	TS-330/TSM, modified with RFL AGC adapter, model HB-16490 for low drive application
Capacity ratio $\frac{C_0}{C_1}$	310 ± 10
Drive level, rated	10 µW
Frequency deviation between -40°C ±0.5°C and upper turning point	Within ± 5 ppm
Crystal quality (Q) 1/	$\geq .5 \times 10^6$ at 30°C
Spurious coupling	The rate of change of crystal resistance with temperature shall not exceed 1 ohm per 3°C over the entire temperature range of -55°C to +90°C.
Capacitance, shunt	3 ±0.5 pF
Shock (specified pulse):	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Vibration, high frequency: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Aging: 3/	
Temperature	65°C ±1°C The instability of temperature control during the aging test shall not exceed ±0.01°C. The average weekly aging rate shall be no greater than .01 ppm per week for the last 2 weeks of a 4-week aging test.

1/ $Q = \frac{1}{4\pi (\Delta f)(C_T)(R)}$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 30 pF.

C_T = Capacitance: C_0 (shunt capacitance of tne crystal unit) plus C_L (load capacitance) of 30 pF

R = Motional-arm resistance of the crystal unit.

2/ Method 204 of MIL-STD-202, test condition A.

3/ The crystal units shall be preconditioned by storage in an oven at 65°C ±5°C for 3 weeks at the time of manufacture.

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© QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE:

NATO guidance type NXT 45.

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - ER, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DLA - ES

User activities:
Army - MI, MU
Navy - AS, MC, SH
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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